

PARAMVEER MULTANI

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EDUCATION

McMaster University

Bachelor of Computer Engineering (Co-op)

Hamilton, Ontario

Expected Graduation Apr 2029

- Relevant Coursework: Algorithm Design and Analysis, Data Structures and Algorithms, Software Development, Artificial Intelligence, Digital System Design, Computer Systems Integration, Microprocessors, Electronic Devices and Circuits

EXPERIENCE

Hoskin Scientific

Instrument Technician Intern

May 2026 – Sep 2026

Oakville, ON

- Built a browser-based offline tool in **JavaScript** and **ExcelJS** that parses instrument CSV exports into calibration worksheets, cutting report prep from **10 minutes to under 1**.
- Pitched the tool to management unprompted; approved and **deployed to all Hoskin offices across Canada**, supporting **20+ calibration reports daily**.
- Performed **PCB-level diagnostics** on **100+** scientific instruments, isolating failed components via multimeter testing and restoring function through resoldering and part replacement.
- Executed lifecycle testing and certification across YSI, DJI, Sontek, and TSI lines, covering calibration, continuity testing, firmware updates, and manufacturer escalation.

BS Transportation

Operations & Technical Assistant

2021 – Present

Hamilton, ON

- Automated **Avaal** freight manifest generation with a **Python** script, cutting per-manifest time from **5 minutes to under 2** and recovering **50+ hours annually**.
- Rebuilt payroll and expense tracking in **Excel**, replacing manual entry and reducing transcription errors across weekly reporting cycles.
- Diagnosed and repaired truck electrical systems by tracing and reorganizing wiring, cutting diagnostic time **25%** and fleet downtime **20%**.

PROJECTS

Catan Coach AI: Game Analysis and Coaching Engine

Personal Project

Jul 2026 – Present

- **Reverse-engineered** colonist.io's undocumented **WebSocket** protocol, decoding **MessagePack** frames across a 113-frame capture to map game state, incremental diffs, and trade events.
- Designing a headless Catan engine in **Python/NumPy** with **expectimax** and **MCTS** search, client-decoupled so one brain serves both live overlay and post-game review.
- Building a **FastAPI** service behind a **TypeScript Manifest V3** extension and **Next.js** app; **PyTorch** value network planned to learn evaluation via **AlphaZero-style self-play**.

Spatial Mapping Using Time-of-Flight Sensor

Personal Project

Jan 2026 – Apr 2026

- Built an embedded **360°** spatial mapping system in **C++** using a **VL53L1X** time-of-flight sensor and stepper motor for indoor distance capture.
- Implemented **I2C** sensor communication and **UART** serial transmission to a PC, reconstructing **3D models** of indoor environments from the data stream.

Automated Baggage Handling System

Team Project

Oct 2024 – Dec 2024

- Prototyped a luggage routing system with a **Q-Arm**, servo table, and barcode scanner for automated classification.
- Wrote **Python** control algorithms achieving **88%** seat detection accuracy and cut runtime errors **30%** through loop optimization.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, MATLAB, R, LaTeX

Frameworks & Tools: Git, GitHub, FastAPI, NumPy, pytest, React, Next.js, Chrome Extensions (Manifest V3), WebSockets, ExcelJS, Linux, VS Code, AutoCAD

Hardware: PCB Diagnostics, Soldering, Multimeter Testing, Arduino, MSP432E401Y, VL53L1X ToF, Quanser, Digilent Analog Discovery, 3D Printing (Prusa)

Interests: Basketball, Chess, Hiking, Hip-Hop